

BRANDON MOSSOP

Ph.D. Candidate, School of Computing, Queen's University

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RESEARCH PROFILE

Research interests: causal artificial intelligence, causal discovery, temporal causal inference, Bayesian machine learning, uncertainty quantification, time-series forecasting, agentic AI, and explainable and robust machine learning.

Research program: Developing methods that learn causal structure from data and use that structure for interpretable forecasting, intervention reasoning, and decision support under non-stationarity and uncertainty.

EDUCATION

Ph.D. in Computing, Queen's University, Kingston, Ontario	Expected 2026
Dissertation: Causal Structure Learning for Robust and Explainable Artificial Intelligence	
Supervisor: Dr. Salimur Choudhury	
M.Sc. in Computer Science, Specialization in Artificial Intelligence, Lakehead University	2022
Thesis: Hybrid Deep Learning with Stacked Dilated Causal Convolutions for Health Forecasting using Multivariate Time-series Data	
Honours B.Sc. in Computer Science, Lakehead University	2020
B.Eng. in Civil Engineering, Lakehead University	2014

PEER-REVIEWED PUBLICATIONS

Journal Articles

1. Mossop, B., & Choudhury, S. (2026). Regime-aware causal Bayesian forecasting for non-stationary time series. PLOS ONE, 21(8), e0356474. <https://doi.org/10.1371/journal.pone.0356474>
2. Mossop, B., & Choudhury, S. (2026). Improving machine learning forecasting explainability and performance with causal feature selection for infectious disease spread. INFOR: Information Systems and Operational Research. Published online August 19, 2026.

Conference Papers

3. Mossop, B., & Choudhury, S. (2025). Agent-guided Causal Discovery with a Small Language Model. 2025 3rd International Conference on Foundation and Large Language Models (FLLM), 693-698. <https://doi.org/10.1109/FLLM67465.2025.11391103>
4. Mossop, B., & Rahman, Q. A. (2023). Infectious Disease Forecasting using Multivariate Incomplete Time-series: A Hybrid Architecture with Stacked Dilated Causal Convolutions. 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 4220-4227. <https://doi.org/10.1109/BIBM58861.2023.10385253>
5. Mossop, B., Bismil, R., & Rahman, Q. A. (2023). Forecasting Hospital Mental-Health Admissions with a Novel Hybrid Deep Learning Architecture. 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), 4093-4100. <https://doi.org/10.1109/BIBM58861.2023.10385786>

Manuscripts / Ongoing Research

Mossop, B., & Choudhury, S. Agentic Causal Discovery with Small Language Models Through Causal Story Reasoning. Manuscript in preparation / under development.

DOCTORAL RESEARCH

Graduate Researcher, GOAL Lab, School of Computing, Queen's University	Ph.D. research
• Develop causal discovery and causal forecasting methods for time-series systems whose relationships change over time.	

- Designed CaReBayes, a regime-aware Bayesian forecasting framework that combines temporal causal discovery, regime identification, Bayesian structural autoregression, and posterior predictive uncertainty.
- Developed causal feature-selection methods for infectious-disease forecasting, using learned causal structure to improve interpretability and predictive performance.
- Investigate agent-guided causal discovery using small language models, including agent roles for orientation, edge augmentation, critique, and causal story reasoning.
- Research emphasizes reproducible experiments, interpretable causal graphs, intervention reasoning, uncertainty quantification, and comparison with statistical and machine-learning baselines.

TEACHING EXPERIENCE

Teaching Assistant, School of Computing, Queen's University

Queen's University

CISC 352: Artificial Intelligence

- Supported instruction in artificial intelligence, including causal reasoning and causal inference concepts.
- Developed and/or delivered teaching material introducing causal graphical models and related concepts within the course.
- Assisted students with technical problem solving, assessment, and course concepts.

SELECTED RESEARCH CONTRIBUTIONS

Causal Forecasting and Non-Stationarity

- Integrates temporal causal structure learning with Bayesian forecasting to model regime-dependent dynamics rather than assuming a fixed data-generating process.
- Uses posterior predictive distributions to quantify forecast uncertainty and supports intervention-oriented forecasting through structural causal models.

Agentic Causal Discovery

- Studies how small language models can contribute contextual knowledge to causal discovery while statistical conditional-independence tests provide data-driven constraints.
- Develops multi-agent workflows for graph orientation, augmentation, criticism, and causal explanation.

Health and Biomedical Forecasting

- Developed deep-learning and causal-ML methods for infectious-disease and hospital-demand forecasting using multivariate time-series data.

PRESENTATIONS AND SCHOLARLY DISSEMINATION

Oral presentation, 3rd International Conference on Foundation and Large Language Models (FLLM)

Vienna, 2025

Agent-guided Causal Discovery with a Small Language Model

IEEE International Conference on Bioinformatics and Biomedicine (BIBM)

2023

Peer-reviewed conference publications in infectious-disease forecasting and hospital mental-health admissions forecasting.

ACADEMIC SERVICE

Technical Program Committee Member, FLLM 2026

2026

Invited service for the International Conference on Foundation and Large Language Models.

TECHNICAL EXPERTISE

Causal inference: structural causal models, DAGs, temporal causal discovery, constraint-based causal discovery, d-separation, interventions, causal feature selection

Bayesian and probabilistic ML: Bayesian regression, posterior predictive inference, uncertainty quantification, probabilistic time-series models, PyMC

Machine learning: deep learning, time-series forecasting, convolutional neural networks, small language models, model evaluation

Programming and research tools: Python, PyTorch, JAX, NumPy, pandas, scikit-learn, Matplotlib, LaTeX, Git/GitHub

SELECTED OPEN RESEARCH SOFTWARE

CaReBayes

GitHub

- Open research repository supporting the regime-aware causal Bayesian forecasting work:
github.com/bdmossop/CaReBayes

REFERENCES

Available upon request. For academic applications, a separate reference list can be supplied with the names, titles, institutions, and contact information of referees.